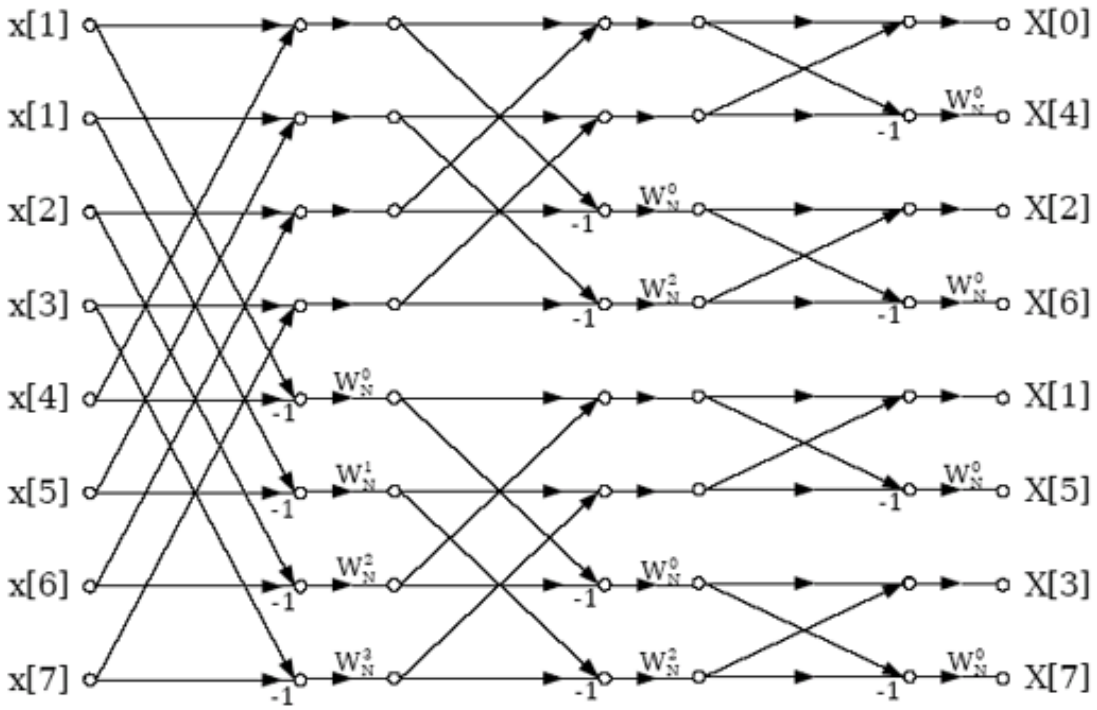
Unit 3 lab : **Advanced Digital Systems Design**

Exercise 1

- Fixed-point architecture for a 512-points FFT
  - 512-point FFT at 62,5MHz according to the algorithm:



Exercise 1

- Steps in the exercise (check the sources folder provided):
  - Study and analysis of the matlab model for the FFT in floating-point
    - *Understand all the computations according to the algorithm*
  - Definition and generation of a suitable input testbench, normalized between +1 and -1, with 12 bits
  - Comparison between the proposed matlab model and *fft m*atlab function
  - Definition of data widths, multipliers and adders, for every operation in the FFT
  - Determination of the quantization error between the fixed-point and the floating-point model
    - *Matlab functions: quantizer and quantize*
    - *Determination of the optimal number of bits for each variable in the architecture, as a trade-off between the accuracy and resource consumption*
    - *Justify properly (using graphs is a suitable way to do it) which is the optimal number of bits*

Exercise 2

- Band-pass filter design:
  - Analyse and design a fixed-point architecture for the filter, following the procedure implemented in the previous exercise
  - Filter characteristic:
    - *FIR*
    - *Sampling frequency 1MHz*
    - *Normalized input between +1 and -1, with 12 bits*
    - *Band-pass definition:*

      fst1= 40e3; fp1= 50e3; fp2= 70e3; fst2= 80e3;

      Ap=0.5; Ast1= 30; Ast2= 30;

  - Determination of the quantization error between the fixed-point and the floating-point model
    - *Matlab functions: fdesign, butter and generatehdl*
    - *Determination of the optimal number of bits for each variable in the architecture, as a trade-off between the accuracy and resource consumption*

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